

```

#empty list
my_list = []
print(my_list)
#list of integers
my_list = [1, 2, 3]
print(my_list)
#multiple data assign
my_list = [1, "Hello", 3.4]
print(my_list)
#nested in dexing
n_list = [2.9, 3000, 'Hello', "Welcome to python list", True]
print(n_list)
n_list = ["Happy", [2, 0, 1, 5]]
print(n_list[0][3])
print(n_list[1][2])
#nested list example
my_list = ["mouse", [8, 4, 6], ['a', 'b', 'c']]
print(my_list)
print(my_list[0])
#nested indexing
print(my_list[1][1])
print(my_list[2][2])
#negative indexing
my_list = ['p', 'r', 'o', 'b', 'e']
print(my_list[-1])
print(my_list[-5])
#appending
odd = [1, 3, 5]
odd.append(7)
print(odd)
#extwend
odd.extend([9, 11, 13])
print(odd)
#concateneting
odd = [1, 3, 5]
print(odd + [9, 7, 5])
print(["re"] * 3)
#list length
thislist = ["apple", "banana", "cherry"]
print(len(thislist))
#remove item
thislist = ["apple", "banana", "cherry"]
thislist.remove("banana")
print(thislist)
#loop list
thislist = ["apple", "banana", "cherry"]
for x in thislist:
    print(x)
#while loop
thislist = ["apple", "banana", "cherry"]
i = 0
while i < len(thislist):
    print(thislist[i])
    i += 1
#sort list
thislist = [100, 50, 65, 82, 23]
thislist.sort()
print(thislist)
#sort list ascending
thislist = ["orange", "mango", "kiwi", "pineapple", "banana"]
thislist.sort()
print(thislist)
#sort list decending

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```

thislist = [100, 50, 65, 82, 23]
thislist.sort(reverse=True)
print(thislist)
thislist = ["orange", "mango", "kiwi", "pineapple", "banana"]
thislist.sort(reverse=True)
print(thislist)
#copy list
thislist = ["apple", "banana", "cherry"]
mylist = thislist.copy()
print(mylist)
#insert
fruits = ['apple', 'banana', 'cherry']
fruits.insert(1, "orange")
print(fruits)
#remove index
thislist = ["apple", "banana", "cherry"]
thislist.pop(1)
print(thislist)
thislist = ["apple", "banana", "cherry"]
thislist.pop()
print(thislist)
thislist = ["apple", "banana", "cherry"]
del thislist[0]
print(thislist)
#clear list
thislist = ["apple", "banana", "cherry"]
thislist.clear()
print(thislist)
#list count() method
fruits = ["apple", "banana", "cherry", "apple"]
x = fruits.count("apple")
print(x)
#list count() method
fruits = [1,4,2,9,7,8,93,1]
x = fruits.count(9)
print(x)
#list method
lst = [10, 20, 30, 40, 50]
value = 30
position = lst.index(value)
print(position)
#built in function
nums = [3, 41, 12, 9, 74, 15]
print(len(nums))
print(max(nums))
print(min(nums))
print(sum(nums))
#split
s = "pining for the fjords"
t = s.split()
print(t)
#list and string
s = "spam"
t = list(s)
print(t)
print(t[2])

```

OUTPUT

```

[]
[1, 2, 3]
[1, 'Hello', 3.4]
[2.9, 3000, 'Hello', 'Welcome to python list', True]

```

```
['mouse', [8, 4, 6], ['a', 'b', 'c']]
mouse
4
c
e
p
[1, 3, 5, 7]
[1, 3, 5, 7, 9, 11, 13]
[1, 3, 5, 9, 7, 5]
['re', 're', 're']
3
['apple', 'cherry']
apple
banana
cherry
apple
banana
cherry
[23, 50, 65, 82, 100]
['banana', 'kiwi', 'mango', 'orange', 'pineapple']
[100, 82, 65, 50, 23]
['pineapple', 'orange', 'mango', 'kiwi', 'banana']
['apple', 'banana', 'cherry']
['apple', 'orange', 'banana', 'cherry']
['apple', 'cherry']
['apple', 'banana']
['banana', 'cherry']
[]
2
1
2
6
74
3
154
['pinning', 'for', 'the', 'fjords']
['s', 'p', 'a', 'm']
a
```